Detecting Machine-Generated Text: How Effective are Current AI-Detection Methods for Non-English Text

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ABSTRACT

Recent advancements in machine-generated text present major challenges to higher education institutes (HEIs). As artificial intelligence (AI) tools such as Google Gemini and ChatGPT become increasingly advanced and accessible worldwide, detecting machine-generated content, particularly in non-English languages remains an underexplored area. This paper aims to fill this gap by investigating how effective current HEI machine-generated text detection tools are in identifying human and AI-generated content in non-English languages. We compare the strengths and limitations of six detection tools based on their performance across five different languages, concluding that current HEI AI detectors are largely ineffective in accurately distinguishing between human and AI-generated text in multiple languages. The study proposes improvements to create a more inclusive and accurate detection model, contributing to the advancement of text detection technology and addressing the global need for more effective research moderation.

1 INTRODUCTION

Higher education institutes (HEIs) are critical in developing professional standards and ethical practices in academic research. With the advancement of large language models (LLMs) such as GPT3, the improvement of AI-generated text on generating unauthorised content has introduced significant challenges to academic integrity. Unauthorised content generation is defined as "*using unapproved technological assistance to produce academic work*" and has become increasingly sophisticated, raising concerns about its impact on assessment integrity (Foltýnek et al. 2023).

While the focus for AI detection has predominantly been on English-based texts as shown in studies by Pan et al (2024) and Webber Wulff et al (2023), the global growing nature of AI technology necessitates an evaluation of detection tools across multiple languages (Pan et al, 2024). The ability of current AI-generated text detection methods to accurately identify machine-generated content in non-English languages remains underexplored. This gap is critical to assess as HEIs worldwide deal with the implications of AI on academic honesty.

This report aims to address this gap by assessing the abilities and accuracy of detection tools for AI-generated text in non-English languages. Specifically, the study investigates:

1. What are the current models for AI and potential cross-language detection.
2. The efficiency and problems of these tools in detecting AI-generated content across various languages.
3. Propose developments for generalised models to detect all languages reliably.

By examining these aspects, the research seeks to explore the effectiveness of detection tools in a multilingual setting and propose improvements for more inclusive and accurate models. The following sections will cover the motivation, review related literature and detail the research methodology, followed by an analysis of findings and proposals for improvements.

AI's rapid progression has transformed many aspects of society, including higher education and scientific research. Large learning models such as ChatGPT can generate text that closely mimics human writing, posing a significant threat to academic integrity (Dwivedi et al, 2023). Although there has been much research done on English-based text, the efficiency of detection tools in recognising AI-generated content in non-English languages has not been extensively studied. This gap is crucial because non-English texts pose distinct linguistic challenges that can impact detection accuracy; For example, Chinese and Japanese texts contain unique logographic script, presenting challenges that are different from those of alphabet-based languages like English. Furthermore, Arabic, with its complex morphology and script, and Spanish, which is spoken in various dialects across Latin America and Spain also add layers of complexity that current detection tools may not adequately address. Additionally, the evolution of AI technology and obfuscation techniques such as paraphrasing, complicates the task of distinguishing between human-written and AI-generated content. According to a recent study by Jemma Skeats et al, approximately 28% of students and academics within an HEI setting admitted to using AI tools for academic purposes in 2024 without being punished, with this number expected to rise as these tools become more accessible and sophisticated (Skeat et al, 2024).

2 BACKGROUND & LITERATURE REVIEW

2.1 Large Language Models & Applications in HEIs

LLMs like GPT-3 have become pivotal in the advancement of natural language processing (NLP). Their ability to generate human-like text has implications across various fields, from automated customer service to academic research (Chiarello et al, 2024). The architecture of most LLMs is based on the Transformer

model. These models use a neural network architecture where the model performs word embedding to convert words into high-dimensional vector representations. Then, the data is passed through multiple transformer layers. Within these layers, the self-attention mechanism plays a crucial role in understanding the relationships between the words in a sequence (Ibrahim, 2023). Finally, the model generates text by predicting the most likely next word in the sequence based on the learned context, which is known as autoregressive language modelling (Yang et al, 2019). LLMs are trained on extensive datasets that cover diverse topics and styles. For instance, GPT-3 was trained on hundreds of billions of words from sources including books, articles, and websites (Ray, 2023). This process helps the model learn the properties of language and generate contextually appropriate text. One of the primary capabilities of LLMs is their ability to generate coherent and contextually relevant text. They can produce essays, articles, and even technical content with remarkable fluency based on user questions (Schulman et al., 2022). The recent advancements and high quality of text generated by LLMs pose significant challenges for detecting AI-generated content. Research has highlighted the difficulty in distinguishing between human-written and AI-generated text, especially when the latter is crafted to mimic specific writing styles (Gehrmann et al, 2019). However, there have been different approaches made in recent years to detect AI-generated text.

2.2 Challenges and Limitations of Current AI-Detectors

Traditional approaches to detect AI-generated text primarily involve statistical outlier methods. These methods utilise different statistical metrics including entropy, perplexity, and n-gram frequency to differentiate between human and AI-generated texts (Gehrmann et al., 2019). An example includes a new innovative statistical detection method created in 2023 called DetectGPT (Mitchell et al., 2023). DetectGPT operates on the principle that text generated by LLMs tends to lie in the negative curvature areas of the model's log probability. It generates AI content and compares multiple differences of model-generated text against the given text to determine whether the text is machine-generated or human-based, according to the log probability as shown in Figure 1. The limitations of statistical detection methods include inaccuracy with evolving LLMs as older statistical methods may struggle to accurately detect newer AI models as they continuously evolve and generate more human-like content. Older models of statistical outlier methods often produce high rates of false positives (misidentifying human text as machine-generated) and false negatives (failing to identify machine-generated text). Furthermore, they struggle with limited adaptability as some detectors may not adapt well to new linguistic patterns or changes in languages other than English. Additionally, handling large volumes of data with statistical methods can be computationally intensive and inefficient (Dwivedi, Y.K. et al, 2023).

Along with statistical methods, classifier-based detectors are also common in natural language detection, particularly in fake news and misinformation detection (Hamed et al, 2023). Classifier methods train models to distinguish human-written text from AI-generated text (Krishna et al, 2023). OpenAI has recently fine-tuned a GPT model using data from Wikipedia, WebText and internal human demonstration data to create a web interface for a detection model using text generated by 34 language models (Elkhatat et al, 2023). This approach combines a classifier-based approach with a human evaluation component to determine whether a given text was machine-generated or human-generated. Similar to traditional approaches, classifier-based detectors have limitations including a high data dependency as these models rely heavily on the quality and diversity of the training data. Accordingly, biases in the training data can lead to inaccurate detections; these detectors also struggle to adapt to newer sophisticated text generation techniques and finally, training and deploying classifier models require significant computational resources, which can be a barrier for large-scale applications (Aldoseri et al, 2023).

An alternative detection method that has significant interest is watermark-based identification. A watermark is an imperceptible modification to the generated text that can be detected post-creation by an algorithm. Effective watermarks are difficult to remove by humans and have little effect on the quality of generated text (Zhao et al, 2023). Recent innovative work by Kirchenbauer et al. (2023) has shown how to incorporate watermarks using only the LLM's logits at each step. The watermarking technique proposed by Kirchenbauer et al involves categorising tokens into "green" and "red" lists to generate distinct patterns. It found that watermarked language models are more likely to select tokens from the green list, based on prior tokens, resulting in watermarks/patterns that are typically unnoticeable to humans (Kirchenbauer, J. et al, 2023). However, the limitations of watermark-based identification as stated by Jiang et al include: The effectiveness of watermarks can be compromised by sophisticated paraphrasing or text alterations which may remove the watermark without significantly altering the content's meaning. There is a risk of generating false positives, where the algorithm mistakenly identifies human-written text as watermarked, especially if the text naturally contains patterns similar to those of the watermarking pattern. Implementing watermarking at scale can be challenging, especially in large-scale language models with diverse applications, where maintaining consistent watermark patterns across all outputs may not be feasible and finally, the algorithm required to detect watermarks must be robust and efficient, which adds computational processing to the overall detection process, slowing down efficiency (Jiang et al, 2024).

2.3 Current AI-Detectors on Non-English Languages

While significant research on AI detection methods has focused on English, foreign texts and English translations present unique linguistic and structural challenges that can complicate detection efforts. Recent studies have started to address this problem, exploring how effective AI-detectors are at detecting machine or human-generated text in non-English.

For instance, a study conducted by Debora Weber Wulf et al. investigated the performance of 12 publicly used AI text detectors including many used in HEIs (such as Turnitin) on many different text types including text that was human-written in a non-English language with a subsequent AI/machine translation to English (Weber-Wulff et al., 2023). Their research revealed a notable drop in accuracy when comparing the detection of human-written texts in English versus those that were translated from non-English languages. As shown in Figure 2, the accuracy for detecting case 01-Hum (Human-written English text) was approximately 96%. However, for texts originally written by humans in non-English languages and then machine-translated to English (Case 02-MT),

the accuracy decreased by 20% (76%). This significant reduction highlights how machine translation can introduce changes that affect detection accuracy, revealing traces of AI in the output even when the original text was completely human written. In addition, the study shows that the current models struggle with obfuscation techniques (06-Para), leading to a 42% accuracy drop (34%). These findings underscore the limitations of current detection methods, particularly when applied to translated non-English texts, and point to a critical need for more refined and language-inclusive detection tools.

2.4 Motivation

Based on the research completed, understanding how well existing tools perform across different languages is essential for developing robust detection methods that can adapt to diverse linguistic contexts. Addressing these limitations is vital for maintaining the credibility of academic assessments and ensuring fair evaluations in a globalised academic environment (Ramesh et al, 2023). This research is motivated by the need to fill this gap in research by assessing the effectiveness of detection tools in identifying AI-generated text across multiple languages. By exploring the accuracy, reliability, and consistency of these tools, the study aims to enhance detection capabilities and support HEIs in managing the challenges posed by AI-generated content.

3. METHODOLOGY

3.1 Dataset & Language Selection

The study utilises a multilingual dataset sourced from XQuAD (Cross-lingual Question Answering Dataset) available on GitHub (Artetxe et al, 2019). This benchmark dataset is specifically designed to evaluate cross-lingual question answering performance and comprises a curated subset of 240 paragraphs and 1,190 question-answer pairs (Rajpurkar et al, 2016).

All question-answer pairs have been professionally translated into 12 different languages, ensuring a high standard of quality and consistency. For this research, the following five specific languages have been chosen to represent a diverse array of linguistic families and structures:

1. English
2. Spanish
3. German
4. Chinese
5. Arabic

3.2 Test Cases

For the first category (named Hum-1), three distinct pieces of human-written text were selected from the dataset for each of the five chosen languages. The selected texts are designed to encompass a diverse range of styles and subject matter, providing a comprehensive assessment of the detection models (See figures 5-9).

Text 1 - Sports Analysis:
This paragraph discusses the performance of the Carolina Panthers defence in the NFL in 2015, including statistics and player contributions.

Text 2 - Historical Analysis:
This information paragraph provides a historical perspective on Warsaw's multiculturalism based on census data.

Text 3 - Theoretical Computer Science
This text outlines concepts in computational complexity theory, characterised by technical terminology.

For the second category (AI-2), the same three topics (sports analysis, historical analysis, and theoretical computer science) were used to generate AI content. This involved employing GPT-4 to produce text based on the questions derived from the dataset. The generated texts were designed to reflect the same complexity and diversity found in the human-written counterparts, allowing for a fair comparison. The AI-generated responses were then compared against the original human-written texts in each language (English, Spanish, German, Chinese and Arabic) to evaluate the effectiveness of detection tools (See figure 10 for detailed AI prompt statement).

3.3 AI-Generator Selection

ChatGPT-4 was selected as the AI text generator for this investigation due to its advanced capabilities and extensive media attention especially in relation to students and its usage. Its proficiency in generating coherent text across multiple languages makes it particularly suitable for examining the effectiveness of AI-generated text detection tools.

Given its ability to handle nuanced language structures and diverse prompts, GPT-4 can produce high-quality outputs that closely mimic human writing. This ensures that the generated texts provide a robust challenge for detection models, ultimately allowing for a comprehensive evaluation of their accuracy and effectiveness in identifying AI-generated content across various languages. Additionally, using a single, well-regarded language model helps standardise the testing process, ensuring that any variations in detection accuracy can be attributed to the detection tools rather than inconsistencies in the generated text.

3.4 AI-Detection Tool Selection

In this investigation, a range of AI detection tools have been selected, tailored for effective analysis of AI-generated text, particularly within the context of HEIs. The tools chosen include:

- DetectGPT (https://detectgpt.com/)
- Go Winston (https://gowinston.ai/)
- GPTZero (https://gptzero.me/)
- GPT-2 Output Detector Demo (https://openai-openai-detector.hf.space/)
- Writer.AI (https://writer.com/ai-content-detector/)
- ZeroGPT (https://www.zerogpt.com/)

These tools were specifically selected for their accessibility and ease of implementation, which is vital for researchers and educators within HEIs who may not have extensive technical expertise. Moreover, their low computational requirements make them particularly advantageous for academic settings where resources may be limited. This ensures that faculty and students can effectively utilise these tools without needing high-end hardware.

The choice of these detection models also aligns with Weber-Wulff et al's research, which highlights the necessity for computationally efficient detection methodologies suitable for educational environments (Weber-Wulff et al, 2023). The recent studies from 2023 reinforce this focus, showcasing detection models that are lightweight and easily deployable in HEIs.

By employing these AI detection tools, the objective is to evaluate their performance in detecting AI-generated text across multiple languages, ultimately contributing to the development of effective strategies for research moderation and academic integrity in higher education contexts.

3.5 Evaluation/Classification Process

For the evaluation, given that the detection tools do not yield exact binary classifications, a classification system was implemented for both categories (Human & AI Written). This system was based on the probabilities reported by the tools, indicating the percentage of the text that was AI-generated, as outlined in Table 1.

Human-written (Negative) text (Hum-1) and the Detector states that it is written by:	
100-80% Artificial Intelligence	False Positive (FP)
80-60% Artificial Intelligence	Unknown False Positive (UFP)
60-40% Artificial Intelligence	Neutral/Unclear (NUC)
40-20% Artificial Intelligence	Unknown True Negative (UTN)
20-0% Artificial Intelligence	True Negative (TN)

AI-written (Positive) text (AI-2) and the detector states that it is written by:	
100-80% Artificial Intelligence	True Positive (TP)
80-60% Artificial Intelligence	Unknown True Positive (UTP)
60-40% Artificial Intelligence	Neutral/Unclear (NUC)
40-20% Artificial Intelligence	Unknown False Negative (UFN)
20-0% Artificial Intelligence	False Negative (FN)

Table 1: Classification of AI-detection percentage for Human-Written & AI-Generated Texts

The collaborative evaluation approach ensured a thorough examination of the detection tools' performance across diverse writing styles and contexts, facilitating a comprehensive understanding of their strengths and limitations in detecting AI-generated content. This approach was not just chosen to ensure thorough examination, but also to provide educators with critical insights into why certain texts were classified as human or AI-written. By reviewing the probability scores using this classification, educators can make informed judgments about the authenticity of submissions, enhancing transparency in the evaluation process. After all the classifications were declared, the results were gathered and the measures of accuracy, precision, false positive rate and false negative rates were calculated.

Accuracy:

When no in-between/unknown results are allowed, i.e. not including UTN, UTP, UNC, UFN and UFP, accuracy is defined as a ratio of correctly classified results to all results.

$$ACC = \frac{TN + TP}{(TN + TP + FN + FP)}$$

Precision

Precision involves finding the ratio of true positives to all positive classification cases.

$$Precision = \frac{TP}{TP + FP}$$

4. RESULTS

To begin, the overall outcomes of the detection tools were evaluated (shown in Table 10-11), categorising them as (unknown) true/false positives and negatives. This classification was assessed based on two key criteria: accuracy/precision and error type. Given the educational context of the study, an error analysis was also performed, as different types of errors hold varying levels of significance.

4.1 Accuracy

Due to the classification also containing semi-correct/unknown results, the regular accuracy formula has to be adjusted to include these results. Within **HEI settings, partial correct classifications** are valuable especially with human editing and AI-paraphrasing being a possibility. Therefore, the second accuracy evaluation will count unknown-correct results being included as correct values. The formula for this is:

$$ACC_{(Incl)} = \frac{TN + TP + UTP + UTN}{(TN + TP + FN + FP + UTP + UTN + UFN + UFP + NUC)}$$

The inclusive accuracy results were then calculated using the results shown in table 2 and 3.

AI-Detector	$Accuracy_{Incl}$
DetectGPT	80.000
Go Winston	73.333
GPT Zero	63.333
GPT-2 Output Detector Demo	90.000
Writer	93.333
ZeroGPT	73.333

Table 4: Calculated inclusive accuracy for each AI-detector

Language	$Accuracy_{Incl}$
English	86.111
Spanish	86.111
German	88.889
Chinese	77.778
Arabic	55.556

Table 5: Calculated inclusive accuracy for each language

Additionally, the regular accuracy was calculated overall for each detector and for each language individually as shown in table 2 and table 3.

It can be seen from tables 2-3 and 5-6, the **accuracy does shift as a result of including the partial results**. Figure 3 and figure 4 present the accuracy for each tool and language as a mean of the two different accuracy approaches along with the two other accuracy approaches.

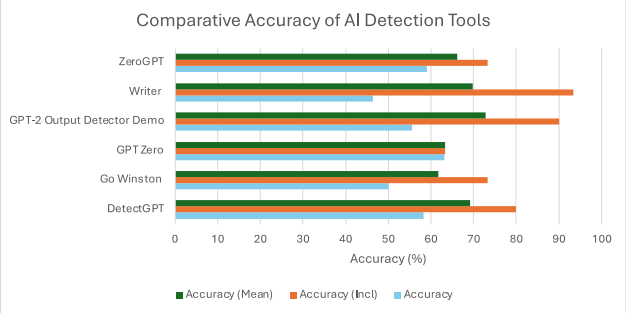


Figure 3: The comparative accuracy of AI detection tools

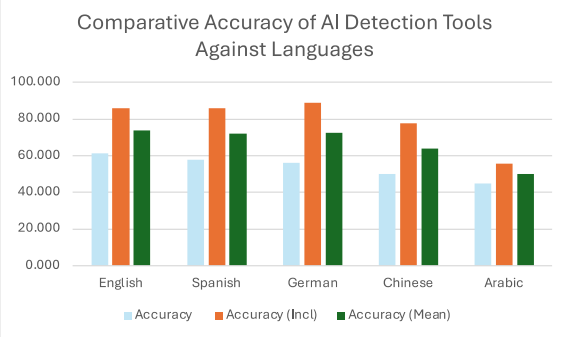


Figure 4: Comparative Accuracy of AI Detectors Across Languages

Overall, the comparative accuracy of English relative to other languages demonstrates only a slight **decrease of 1.613% for Spanish**, indicating that the detection tools maintain a consistent level of performance across these two languages. However, there is a **more pronounced decline** in accuracy for languages that **do not utilise the Latin alphabet with a drop of 9.812% for Chinese and a staggering 23.423% for Arabic**. These figures suggest that the inherent complexities of non-Latin scripts pose significant challenges for AI detection tools, highlighting the importance of script familiarity in the development and training of these models. Interestingly, German showed a 1.131% increase in accuracy, suggesting that the detection tools may perform better with certain Germanic languages compared to their Latin counterparts. This improvement could be attributed to the structural similarities between German and English, which **may assist the detection process.**

In terms of the tools themselves, the GPT-2 Output Detector Demo achieved the highest mean accuracy at 72.778% and inclusive accuracy of 90%, indicating that it is the most reliable tool among those tested. This performance suggests that its algorithm is particularly effective at distinguishing between human-written and AI-generated content. Finally, Go Winston recorded the lowest average accuracy at 61.667%, raising concerns about its effectiveness especially in academic and educational settings where precision is crucial.

4.2 Precision

Once more, unknown true/false positives had to be evaluated into the formula. Again, the inclusive formula will count them as positively classified and therefore precise:

$$Precision_{(Incl)} = \frac{TP + UTP}{TP + FP + UTP + UFP}$$

Language	Precision	Precision (Incl)
English	60.000	66.667
Spanish	60.000	60.000
German	62.500	60.000
Chinese	50.000	55.556
Arabic	44.444	50.000

Table 6: Calculated Precision & Inclusive Precision (%) for Each Language

AI-Detector	Precision	Precision (Incl)
DetectGPT	50.000	57.143
Go Winston	47.059	52.381
GPT Zero	80.000	85.714
GPT-2 Output Detector Demo	66.667	66.667
Writer	0.000	0.000
ZeroGPT	60.000	58.333

Table 7: Calculated precision & inclusive precision (%) for each AI-Detector

The precision percentage presented in table 6 highlights significant variability in the performance of AI detection tools across different languages. English demonstrates a precision rate of 60%, with an inclusive precision rising to 66.667%, indicating a relatively strong performance in detecting AI-generated content. Similarly, Spanish achieves a precision of 60%, though its inclusive precision remains unchanged, suggesting consistent performance across both measures. German displays the highest precision score of 62.500%, which increases to 66.667% when unknown classifications are considered. However, performance declines for non-Latin script languages. Chinese records a lower precision of 50%, with a modest improvement to 55.556% in inclusive precision. Arabic, meanwhile, shows the lowest precision at 44.444%, with an inclusive precision of 50%. These results suggest that AI detection tools are more effective when applied to languages that use a Latin alphabet compared to non-Latin languages such as Chinese and Arabic.

Table 7 provides a detailed evaluation of the precision scores for each AI detection tool. Notably, GPT Zero was found to be the most reliable tool, achieving a precision of 80% with an inclusive precision of 85.714%. This indicates its strong capability in accurately identifying AI-generated content. The GPT-2 Output Detector Demo also demonstrates a relatively high precision score of 66.667%. However, it is worth highlighting that this represents a significant drop from its inclusive accuracy score of 90%, suggesting that **while the tool may excel in general accuracy, its precision, particularly its ability to correctly identify true positives requires further refinement.**

Other tools display varying degrees of performance. DetectGPT records a precision score of 50%, with an inclusive precision of 57.143%, while Go Winston shows the lowest precision at 47.059%, which marginally improves to 52.381% when inclusive precision is considered. Writer.AI, however, does not detect any AI-generated content accurately, resulting in a precision score of 0%. ZeroGPT, in contrast, delivers a consistent performance with a precision score of 60%, and a slight increase to 58.333% when including partial results.

The overall results indicate that the tools vary considerably in their precision, with some tools demonstrating higher reliability in detecting AI-generated content, while others show limitations, particularly when handling more complex languages.

4.3 Analysis of Errors (False Positives & False Negatives)

In this section, two types of classification errors that could have significant consequences in HEI settings are being analysed accordingly to the different language detections: false positives leading to false accusations of AI usage against a student and false negatives allowing students to use AI without detected and dealt with in a proper manner.

4.3.1 False Positives: Unfair Student Punishment

False Positives refer to the incorrect detection of AI usage within a human-written piece of text and represent a particularly harmful outcome in academic settings. When a HEI detection tool mistakenly flags a human-written work as AI-generated, the consequences can be severe especially for students. A false positive can lead to unfair accusations of academic dishonesty, resulting in potential penalties such as lower grades, academic punishment or even expulsion.

Given these possible consequences, it is crucial to understand the occurrence of false positives across different languages in detection. The total number and percentage of false positives detected during testing can be found and calculated using the data from Table 3, which outlines the outcomes of each tool across English, Spanish, German, Chinese, and Arabic texts.

	FP	FP (%)
English	6	16.67
Spanish	4	11.11
German	3	8.33
Chinese	7	19.44
Arabic	5	13.89

Table 8: Total false positives & percentage of false positives against the total tests.

Table 8 illustrates the total number of False Positives identified for each language, along with the corresponding percentage of false positives relative to the total tests conducted. From the data, Chinese recorded the highest number of false positives at 7, leading to a FP percentage of 19.44%. This was followed by English, which had 6 false positives, translating to 16.67%. Arabic also presented a notable FP rate of 13.89% with 5 instances. Spanish had 4 false positives, amounting to 11.11%, while German demonstrated the lowest occurrence with 3 false positives, resulting in a percentage of 8.33%.

The findings highlight the varying degrees of reliability among detection tools, particularly when assessing texts written in different languages. The higher false positive rates observed in Chinese and English indicate potential weaknesses in these tools' ability to accurately distinguish between human-written and AI-generated content, thereby necessitating further improvements to minimise the risk of penalising students who have committed no wrongdoings.

4.3.2 False Negatives: Advantageous Usage of AI

False Negatives (FN) occur when AI-generated text is incorrectly classified as human-written, potentially providing certain students with an unfair advantage in academic settings. This misclassification can undermine the integrity of educational assessments, as it allows students who may have utilised AI assistance to present their work without proper attribution or acknowledgment.

	FN	FN (%)
English	6	16.67
Spanish	9	25.00
German	11	30.56
Chinese	7	19.44
Arabic	6	16.67

Table 9: Total false negatives (FN) & percentage of false negatives against the total tests.

Table 9 displays the total number of False Negatives detected across the various languages, alongside the corresponding percentage of false negatives relative to the total tests conducted. The results indicate that Spanish has the highest FN count at 9, resulting in a FN percentage of 25.00%. German follows closely with 11 false negatives, reflecting a FN percentage of 30.56%. English and Arabic both demonstrate similar FN rates of 16.67% and 6 occurrences, while Chinese has 7 false negatives, yielding a FN percentage of 19.44%. These figures highlight the varying effectiveness of detection tools in distinguishing between human and AI-generated content, particularly in languages like German and Spanish, where the misclassification rates are alarmingly high.

The implications of high false negative rates are concerning for academic integrity, as students may exploit the issues of detection tools to submit work that does not reflect their own academic efforts. Addressing this issue is crucial for ensuring fairness in educational environments, emphasising the need for improved detection methodologies that can effectively discern AI-generated text across multiple languages.

5. DISCUSSION

The experimental results reveal that while AI detection tools exhibit decent accuracy across various languages, **their precision, especially for non-English languages, is insufficient** with all precision scores remaining below 70%. This imprecision can be attributed to several factors, including the inherent linguistic diversity and complexity of different scripts. These challenges are particularly pronounced in **non-Latin languages** like Chinese and Arabic, where accuracy dropped by 9.812% and 23.423%, respectively.

A key reason for this disparity lies in the **structural differences between Latin and non-Latin scripts**. AI detection tools are primarily trained on large English or Latin-alphabet datasets, making it easier for these models to detect patterns in languages like English, German, and Spanish. Latin-based languages share syntactic and grammatical similarities that help detection tools recognise AI-generated patterns more effectively. For instance, the slight increase in accuracy for German (+1.131%) suggests that tools trained on English may transfer some knowledge to German due to their similar linguistic structures. The same cannot be said for languages like Chinese or Arabic, which have unique character sets.

Chinese uses logographic characters, where individual symbols represent words, unlike Latin-alphabetic systems. This presents a significant challenge for AI detection models, which may be better suited to recognising alphabetic structures. Additionally, Arabic employs a unique system of diacritics and has a script that reads right to left (Alshammari et al, 2024). These differences have been found to confuse AI detection models that have been optimised for left-to-right reading scripts or languages without diacritical marks, contributing to a lower accuracy and precision. **The higher incidence of False Positives in non-Latin scripts is particularly concerning in academic settings**, as it risks penalising students who have written their own work. Meanwhile, False Negatives allow for AI-generated content to pass undetected, undermining the goal of preventing academic dishonesty.

Furthermore, **another issue with current AI detectors is the ambiguity in detection results**, such as a score of 40-80%, which can create significant challenges in higher education institutions. These unclear classifications make it difficult for educators and administrators to interpret whether a piece of text is human or AI generated. The consequences of these misclassifications can be severe affecting students' grades and reputations, as well as the institution's trust in detection technologies.

To improve the detection of AI-generated content across various linguistic structures, new generalised models must adopt various strategies to create a **flexible, inclusive system**. One key approach is the development of a **hierarchical classifier system** that can first identify the script type of the input text before applying a tailored classifier for that specific language group. By doing so, detection models would be better equipped to handle the unique linguistic patterns present in non-Latin languages like Arabic or Chinese. For example, after determining the script type, the model could employ specialised techniques designed to enhance detection accuracy, ensuring efficient operation across diverse languages and handling their complexities more effectively.

To complement this structural adaptation, the model would require **high-quality, multilingual training data** to achieve true generalisation. Many existing HEI models are primarily trained on English or Latin-alphabet datasets, which limits their effectiveness in processing non-Latin scripts. By expanding the training data to include languages with complex grammatical structures and scripts, such as Turkish or Arabic, the model would be more inclusive, enhancing its ability to detect AI-generated content in underrepresented languages. Such inclusivity is crucial for developing a model that generalises well across various linguistic landscapes, addressing the current imbalance in AI detection models used by higher education institutions. Current literature has increasingly emphasised the importance of training AI detection models on diverse linguistic data. A 2023 paper by Liang, W. et al suggested that the inclusion of non-English research during model training is pivotal to improving the biases against non-English text. The results within this study support that notion and highlight how Latin-script languages such as Spanish and German maintain

higher accuracy levels, while non-Latin languages experience a significant decline.

Additionally, **preprocessing techniques should be tailored to specific language features to further improve the model's generalisation**. Non-Latin scripts such as Arabic, which uses diacritics and Chinese with its logographic characters can pose challenges for AI models. For instance, stripping diacritics from Arabic text or applying segmentation techniques to Chinese characters can simplify the input, allowing the model to focus on deeper linguistic structures. These steps would help reduce misclassifications, especially in the form of false positives and false negatives, leading to more accurate results across various languages.

Incorporating embedding techniques like Multilingual BERT (mBERT) could significantly enhance the detection of AI-generated text across multiple languages by representing them in a shared vector space (Devlin et al., 2018). This approach enables the model to recognise differing linguistic patterns more effectively, especially when dealing with complex scripts such as Arabic, which is read right to left. An example of this includes a study conducted on language detection using the pretrained mBERT model which demonstrates its effectiveness. This study involved creating a language detection model using the Wikipedia 40 billion Multilingual Dataset, which contains entries from 41 different languages. The model was specifically trained on 16 of these languages and mBERT, showcasing the potential of large, diverse datasets for developing robust multilingual detection systems (Kithannae, 2021). Current detection tools such as DetectGPT and GPT-2 Output Detector often struggle with non-English text because they lack the multilingual adaptability that embeddings provide. This capability enhances cross-linguistic detection, making it possible to identify AI-generated content more consistently and accurately across diverse languages and scripts.

Lastly, **confidence thresholds** can be integrated to improve decision-making in AI detection, especially in academic environments like HEIs, where misclassifications can have serious consequences. By setting clear thresholds (e.g. 85% and above for AI-generated content), users can interpret results more easily. Combined with **human-in-the-loop** systems, this approach can ensure that content flagged by the AI is reviewed by human experts, thus combining the efficiency of automated detection with human oversight. This would help in reducing errors and ensuring that the detection process is reliable and transparent in higher education settings.

By adopting these strategies, a generalised HEI AI detection model can be developed that handles the complexities of non-Latin scripts and reduces misclassification risks. Such a model would provide a more inclusive and fair solution, particularly in higher education institutions, where the diversity of language demands a more adaptable and reliable detection system.

6. CONCLUSION & FUTURE WORK

The findings of this paper highlight the significant limitations of current HEI AI text detection tools and their inability to fairly and effectively detect human or AI-generated text across multiple languages. Throughout the experiments, these detection tools frequently presented false positives and false negatives especially for non-Latin scripts, which could lead to incorrect punishments for genuine work or a lack of appropriate action for AI-generated content. These inaccuracies display the need for more reliable and effective detection tools, especially in multilingual contexts.

To enhance the efficiency of future AI detection models, a hierarchical model that first identifies the script type before applying a tailored classifier may significantly improve detection accuracy across various languages. Furthermore, training these models on high-quality multilingual datasets will enable them to effectively address the grammatical and structural differences apparent in languages such as Chinese and Arabic. Implementing specialised preprocessing techniques such mBERT could also reduce misclassifications.

Moreover, integrating clearly defined confidence thresholds would enhance the interpretability of results for educators, thereby facilitating more informed decision-making. By adopting these strategies, we can significantly bolster the reliability and effectiveness of AI detection models, ensuring their capacity to accommodate the diverse linguistic contexts encountered in real-world applications.

Future work should focus on testing and comparing specialised detectors with generalised models to assess whether they can provide better accuracy and reduce the risk of unjust consequences. A key area of exploration is the fine-tuning of AI detection models for underrepresented languages. Special attention should be given to minimising False Positives (FPs) and False Negatives (FNs) to ensure fairness in detecting both human and AI-generated content across diverse linguistic landscapes.

Additionally, future studies should also take into account the fact that the human-written text used in this research was exposed to the internet. Since much of the training data for LLMs originates from online material, there is the potential for AI detection tools to have encountered similar content during training. Future research should focus on testing detection tools using data that is not publicly available online and thus not likely to be included in LLM training datasets.

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APPENDIX

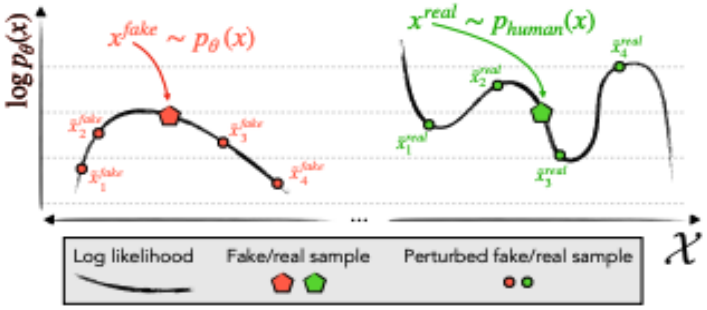


Figure 1: Displays DetectGPT and the tendency of machine-generated text $x \sim p_\theta(x)$ (left) to lie in negative curvature regions of $\log p_\theta(x)$, where the created samples (red samples) have lower model log probability on average identifying AI-generated text. In contrast, human-written text $x \sim p_{human}(x)$ (right) tends not to have a clear negative log probability curvature with nearby samples (green samples) may have both higher and lower log probability (Mitchell et al, 2023).

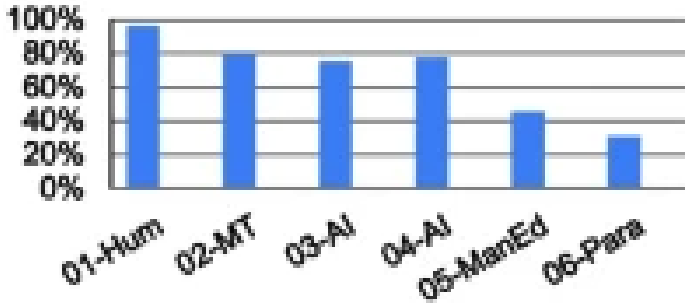


Figure 2: The logarithmic accuracy for each document type by all detection tool for different AI or human-written text. (Weber-Wulff et al., 2023)

English (Human Written)

1. "The Panthers defence gave up just 308 points, ranking sixth in the league, while also leading the NFL in interceptions with 24 and boasting four Pro Bowl selections. Pro Bowl defensive tackle Kawann Short led the team in sacks with 11, while also forcing three fumbles and recovering two. Fellow lineman Mario Addison added 6 sacks. The Panthers line also featured veteran defensive end Jared Allen, a 5-time pro bowler who was the NFL's active career sack leader with 136, along with defensive end Kony Ealy, who had 5 sacks in just 9 starts. Behind them, two of the Panthers three starting linebackers were also selected to play in the Pro Bowl: Thomas Davis and Luke Kuechly. Davis compiled 5 sacks, four forced fumbles, and four interceptions, while Kuechly led the team in tackles (118) forced two fumbles and intercepted four passes of his own. Carolina's secondary featured Pro Bowl safety Kurt Coleman, who led the team with a career high seven interceptions, while also racking up 88 tackles and Pro Bowl cornerback Josh Norman, who developed into a shutdown corner during the season and had four interceptions, two of which were returned for touchdowns.",

2. "Throughout its existence, Warsaw has been a multi-cultural city. According to the 1901 census, out of 711,988 inhabitants 56.2% were Catholics, 35.7% Jews, 5% Greek orthodox Christians and 2.8% Protestants. Eight years later, in 1909, there were 281,754 Jews (36.9%), 18,189 Protestants (2.4%) and 2,818 Mariavites (0.4%). This led to construction of hundreds of places of religious worship in all parts of the town. Most of them were destroyed in the aftermath of the Warsaw Uprising of 1944. After the war, the new communist authorities of Poland discouraged church construction and only a small number were rebuilt.",

3. "Computational complexity theory is a branch of the theory of computation in theoretical computer science that focuses on classifying computational problems according to their inherent difficulty and relating those classes to each other. A computational problem is understood to be a task that is in principle amenable to being solved by a computer, which is equivalent to stating that the problem may be solved by mechanical application of mathematical steps, such as an algorithm. When considering computational problems, a problem instance is a string over an alphabet. Usually, the alphabet is taken to be the binary alphabet (i.e., the set {0,1}), and thus the strings are bitstrings. As in a real-world computer, mathematical objects other than bitstrings must be suitably encoded. For example, integers can be represented in binary notation, and graphs can be encoded directly via their adjacency matrices, or by encoding their adjacency lists in binary"

English: AI Written

1. The Panthers' defence had a standout season, allowing just 308 points, the sixth-best in the league, and leading the NFL with 24 interceptions. They had four Pro Bowl selections, including defensive tackle Kawann Short, who led the team with 11 sacks, forced three fumbles, and recovered two. Defensive lineman Mario Addison contributed 6.5 sacks, while veteran Jared Allen, a five-time Pro Bowler and the NFL's active career sack leader with 136, anchored the line alongside Kony Ealy, who recorded 5 sacks in 9 starts. At linebacker, Thomas Davis and Luke Kuechly were both Pro Bowl selections. Davis registered 5.5 sacks, forced four fumbles, and picked off four passes, while Kuechly led the team with 118 tackles, forced two fumbles, and had four interceptions. In the secondary, Pro Bowl safety Kurt Coleman had a career-high seven interceptions and 88 tackles, while Pro Bowl cornerback Josh Norman emerged as a shutdown corner with four interceptions, returning two for touchdowns.
2. Warsaw has long been a multicultural city. According to the 1901 census, the city had 711,988 residents, of whom 56.2% were Catholics, 35.7% were Jews, 5% were Greek Orthodox Christians, and 2.8% were Protestants. By 1909, the population included 281,754 Jews (36.9%), 18,189 Protestants (2.4%), and 2,818 Mariavites (0.4%). This diverse population led to the construction of numerous places of worship throughout the city. However, most of these were destroyed during the Warsaw Uprising in 1944. After the war, the communist authorities in Poland discouraged the rebuilding of churches, and only a few were restored.
3. Computational complexity theory is a field within theoretical computer science that focuses on categorizing computational problems based on their intrinsic difficulty and examining how these categories relate to one another. A computational problem refers to a task that can theoretically be solved by a computer, which means it can be addressed by applying a series of mathematical steps, such as an algorithm. In the context of computational problems, an instance is represented as a string composed of symbols from an alphabet. Typically, the alphabet consists of binary digits (i.e., the set {0,1}), so the strings are bitstrings. Mathematical entities beyond bitstrings, such as integers or graphs, need to be encoded appropriately. For example, integers are commonly represented in binary form, while graphs may be encoded either through their adjacency matrices or by converting their adjacency lists into binary.

Figure 5: Displays the Human (Left) & AI-generated (Right) text in English

Spanish (Human Written)

1. Los Panthers, que además de liderar las intercepciones de la NFL con 24 y contar con cuatro jugadores de la Pro Bowl, cedieron solo 308 puntos en defensa y se sitúan en el sexto lugar de la liga. Kawann Short, tacle defensivo de la Pro Bowl, lideró al equipo con 11 capturas, 3 balones sueltos forzados y 2 recuperaciones. A su vez, el liniero Mario Addison, consiguió 6 capturas y media. En la línea de los Panthers, también destacó como ala defensiva el veterano Jared Allen —5 veces jugador de la Pro Bowl y que fue el líder, en activo, de capturas de la NFL con 136— junto con el también ala defensiva Kony Ealy, que lleva 5 capturas en solo 9 partidos como titular. Detrás de ellos, Thomas Davis y Luke Kuechly, dos de los tres apoyadores titulares que también han sido seleccionados para jugar la Pro Bowl. Davis se hizo con 5 capturas y media, 4 balones sueltos forzados y 4 intercepciones, mientras que Kuechly lideró al equipo en derribos (118), forzó 2 balones sueltos e interceptó 4 pases. La secundaria de Carolina contó, por un lado, con la seguridad del jugador de la Pro Bowl Kurt Coleman que asumió las riendas del equipo gracias a sus 7 intercepciones (nunca había conseguido tantas hasta ahora) y a sus 88 derribos, y, por otro lado, con el esquinero Josh Norman, también jugador de la Pro Bowl y que a pesar de haber estado de capa caída durante la temporada, consiguió 4 intercepciones, de las cuales dos se convirtieron en touchdowns."

2. "A lo largo de su existencia, Varsovia siempre ha sido una ciudad multicultural. Según el censo del año 1901, de sus 711 988 habitantes, el 56,2 % eran católicos, el 35,7 % judíos, el 5 % cristianos ortodoxos y el 2,8 % protestantes. Ocho años más tarde, en 1909, habían 281 754 judíos (36,9 %), 18 189 protestantes (2,4 %) y 2818 mariavitas (0,4 %). Esto llevó a la construcción, por toda la ciudad, de cientos de lugares de culto religioso; la mayoría de los cuales fueron destruidos tras el Alzamiento de Varsovia en 1944. Al finalizar la guerra, las nuevas autoridades comunistas de Polonia desincentivaron la construcción de iglesias y muy pocas fueron reconstruidas.",

3. La teoría de la complejidad computacional es una rama de la teoría de la computación dentro de la ciencia computacional teórica. Esta teoría se centra en la clasificación de los problemas computacionales según su dificultad inherente y en la relación de esas clases de dificultad entre sí. Un problema computacional se entiende como una tarea que en principio puede ser resuelta por un ordenador, es decir, que el problema puede ser resuelto mediante la aplicación mecánica de pasos matemáticos tales como un algoritmo.",

Spanish (AI Translated)

1. Los Panthers, que lideraron la NFL en intercepciones con 24 y contaron con cuatro jugadores seleccionados al Pro Bowl, cedieron solo 308 puntos en defensa, ubicándose en el sexto lugar de la liga. El tackle defensivo Kawann Short, seleccionado al Pro Bowl, lideró al equipo con 11 capturas, 3 balones sueltos forzados y 2 recuperaciones. Por su parte, el liniero Mario Addison sumó 6 capturas y media. En la línea defensiva, también destacaron el veterano ala defensiva Jared Allen —cinco veces jugador del Pro Bowl y líder activo de capturas de la NFL con 136— junto con el ala defensiva Kony Ealy, quien logró 5 capturas en solo 9 partidos como titular. Detrás de ellos, se encontraban Thomas Davis y Luke Kuechly, dos de los tres apoyadores titulares que también fueron seleccionados para el Pro Bowl. Davis acumuló 5 capturas y media, 4 balones sueltos forzados y 4 intercepciones, mientras que Kuechly lideró al equipo en derribos con 118, forzó 2 balones sueltos e interceptó 4 pases. En la secundaria, Carolina contó con la solidez del safety Pro Bowler Kurt Coleman, quien lideró al equipo con un récord personal de 7 intercepciones y registró 88 derribos, y con el esquinero Josh Norman, también seleccionado al Pro Bowl, quien se consolidó como un esquinero de élite con 4 intercepciones, de las cuales dos fueron devueltas para touchdown.

2. A lo largo de su historia, Varsovia ha sido siempre una ciudad multicultural. Según el censo de 1901, de sus 711,988 habitantes, el 56.2 % eran católicos, el 35.7 % judíos, el 5 % cristianos ortodoxos y el 2.8 % protestantes. En 1909, ocho años después, había 281,754 judíos (36.9 %), 18,189 protestantes (2.4 %) y 2,818 mariavitas (0.4 %). Esta diversidad religiosa impulsó la construcción de cientos de lugares de culto en toda la ciudad, muchos de los cuales fueron destruidos durante el Alzamiento de Varsovia en 1944. Tras la guerra, las nuevas autoridades comunistas de Polonia desalentaron la construcción de iglesias, y solo unas pocas fueron reconstruidas.

3. La teoría de la complejidad computacional es un campo dentro de la informática teórica que se enfoca en categorizar los problemas computacionales según su dificultad intrínseca y examinar cómo se relacionan estas categorías entre sí. Un problema computacional se refiere a una tarea que, en teoría, puede ser resuelta por una computadora, lo que significa que puede abordarse aplicando una serie de pasos matemáticos, como un algoritmo. En el contexto de los problemas computacionales, una instancia se representa como una cadena compuesta por símbolos de un alfabeto. Típicamente, el alfabeto consiste en dígitos binarios (es decir, el conjunto {0,1}), por lo que las cadenas son cadenas de bits. Las entidades matemáticas más allá de las cadenas de bits, como los enteros o los gráficos, deben ser codificadas de manera adecuada. Por ejemplo, los enteros se representan comúnmente en forma binaria, mientras que los gráficos pueden codificarse a través de sus matrices de adyacencia o convirtiendo sus listas de adyacencia en binario.

Figure 6: Displays the Human (Left) & AI-generated (Right) text in Spanish

German (Human Written)

1. Die Verteidigung der Panthers gab nur 308 Punkte ab und belegte den sechsten Platz in der Liga, während sie die NFL mit 24 Interceptions in dieser Kategorie anführte und sich mit vier Pro Bowl-Selektionen rühmen konnte. Pro Bowl Defensive Tackle Kawann Short führte das Team mit 11 Sacks an, erzwang zudem drei Fumbles und erzielte zwei Fumble Recoverys. Mario Addison, ebenfalls Lineman, addierte 6½ Sacks hinzu. Die Panthers-Line präsentierte auch den erfahrenen Defensive End Jared Allen, einen 5-fachen Pro-Bowler, der mit 136 Sacks der aktive Anführer in der NFL-Kategorie Karriere-Sacks war, sowie den Defensive End Kony Ealy, der 5 Sacks in nur 9 Starts erzielte. Nach ihnen wurden zwei der drei Linebacker der Panthers ausgewählt, um im Pro Bowl zu spielen: Thomas Davis und Luke Kuechly. Davis erzielte 5½ Sacks, vier erzwungene Fumbles und vier Interceptions, während Kuechly das Team bei den Tackles anführte (118), zwei Fumbles erzwang und vier Pässe abfing. Carolinas Secondarys bestanden aus dem Pro Bowl-Safety Kurt Coleman, der das Team mit einem Karrierehoch von sieben Interceptions anführte und gleichzeitig 88 Tackles erzielen konnte, und Pro Bowl-Cornerback Josh Norman, der sich während der Saison zur Shutdown Corner entwickelte und vier Interceptions erzielte, von denen zwei zu Touchdowns für sein Team wurden.

2. Warschau war während seiner gesamten Existenz eine multikulturelle Stadt. Laut der Volkszählung von 1901 waren von 711.988 Einwohnern 56,2% Katholiken, 35,7% Juden, 5% griechisch-orthodoxe Christen und 2,8% Protestanten. Acht Jahre später, im Jahr 1909, gab es 281.754 Juden (36,9%), 18.189 Protestanten (2,4%) und 2.818 Mariaviten (0,4%). Dies führte zum Bau von Hunderten von religiösen Kultstätten in allen Teilen der Stadt. Die meisten von ihnen wurden nach dem Warschauer Aufstand von 1944 zerstört. Nach dem Krieg rieten die neuen kommunistischen Behörden Polens vom Kirchenbau ab und nur wenige wurden wiederaufgebaut.

3. Die Komplexitätstheorie ist ein Zweig der Berechnungstheorie in der theoretischen Informatik, der sich darauf konzentriert, Rechenprobleme nach ihrer inhärenten Schwierigkeit zu klassifizieren und diese Klassen miteinander in Beziehung zu setzen. Unter einem Rechenproblem versteht man eine Aufgabe, die grundsätzlich von einem Computer gelöst werden kann. Dies ist gleichbedeutend mit der Aussage, dass das Problem durch die mechanische Anwendung mathematischer Schritte, beispielsweise durch einen Algorithmus, gelöst werden kann.

German: AI CONTENT

1. Die Verteidigung der Panthers ließ nur 308 Punkte zu und belegte damit den sechsten Platz in der Liga, während sie auch die NFL mit 24 Interceptions anführte und vier Pro Bowl-Auswahlen vorweisen konnte. Pro Bowl-Defensive Tackle Kawann Short führte das Team mit 11 Sacks an und erzwang außerdem drei Fumbles und eroberte zwei. Sein Kollege Mario Addison fügte 6 Sacks hinzu. Die Verteidigungslinie der Panthers wurde außerdem von dem erfahrenen Defensive End Jared Allen verstärkt, einem fünfmaligen Pro Bowler, der mit 136 Sacks der führende Spieler der Liga in dieser Kategorie war, sowie von Defensive End Kony Ealy, der in nur 9 Starts 5 Sacks erzielte. Dahinter wurden zwei der drei Starting-Linebacker der Panthers ebenfalls in den Pro Bowl gewählt: Thomas Davis und Luke Kuechly. Davis sammelte 5 Sacks, erzwang vier Fumbles und fing vier Interceptions, während Kuechly das Team mit 118 Tackles anführte, zwei Fumbles erzwang und selbst vier Pässe abfing. Die Secondary der Panthers wurde von Pro Bowl-Safety Kurt Coleman angeführt, der das Team mit einer Karrierebestleistung von sieben Interceptions anführte und zudem 88 Tackles verzeichnete, während Pro Bowl-Cornerback Josh Norman sich zu einem Shutdown-Corner entwickelte und vier Interceptions fing, von denen er zwei zu Touchdowns zurücktrug.

2. Im Laufe seiner Geschichte war Warschau eine multikulturelle Stadt. Laut der Volkszählung von 1901 waren von den 711.988 Einwohnern 56,2 % Katholiken, 35,7 % Juden, 5 % griechisch-orthodoxe Christen und 2,8 % Protestanten. Acht Jahre später, im Jahr 1909, gab es 281.754 Juden (36,9 %), 18.189 Protestanten (2,4 %) und 2.818 Mariaviten (0,4 %). Dies führte zum Bau von Hunderten religiösen Gebetshäusern in allen Teilen der Stadt. Die meisten von ihnen wurden jedoch nach dem Warschauer Aufstand von 1944 zerstört. Nach dem Krieg entmutigten die neuen kommunistischen Behörden Polens den Bau von Kirchen, und nur eine kleine Zahl wurde wieder aufgebaut.

3. Die Theorie der Berechenbarkeitskomplexität ist ein Zweig der Theoretischen Informatik, der sich auf die Klassifizierung von Rechenproblemen nach ihrem inhärenten Schwierigkeitsgrad konzentriert und diese Klassen miteinander in Beziehung setzt. Ein Rechenproblem wird als eine Aufgabe verstanden, die prinzipiell von einem Computer gelöst werden kann, was gleichbedeutend damit ist, dass das Problem durch die mechanische Anwendung mathematischer Schritte, wie eines Algorithmus, gelöst werden kann. Bei der Betrachtung von Rechenproblemen ist ein Problembeispiel eine Zeichenkette über einem Alphabet. In der Regel wird das Alphabet als das binäre Alphabet (d. h. die Menge {0,1}) angenommen, sodass die Zeichenketten Bitstrings sind. Wie in einem realen Computer müssen andere mathematische Objekte als Bitstrings entsprechend codiert werden. Beispielsweise können ganze Zahlen in Binärdarstellung codiert werden, und Graphen können entweder direkt über ihre Adjazenzmatrizen oder durch die Codierung ihrer Adjazenzlisten in Binärform dargestellt werden.

Figure 7: Displays the Human (Left) & AI-generated (Right) text in German

Chinese (Human Written)

1. 黑豹队的防守只丢了 308分，在联赛中排名第六，同时也以 24 次拦截领先国家橄榄球联盟 (NFL)，并且四次入选职业碗。职业碗防守截锋卡万·肖特以 11 分领先于全队，同时还有三次迫使掉球和两次重新接球。他的队友马里奥·爱迪生贡献了 6½ 次擒杀。黑豹队的防线上有经验丰富的防守端锋贾里德·艾伦，他是五次职业碗选手，曾以 136 次擒杀成为 NFL 职业生涯中的活跃领袖。另外还有在 9 场首发中就拿下 5 次擒杀的防守端锋科尼·伊利。在他们身后，黑豹队的三名首发线卫中有两人入选了职业碗：托马斯·戴维斯和卢克·坎克利。戴维斯完成了 5½ 次擒杀、四次迫使掉球和四次拦截，而坎克利带领球队在擒抱 (118) 中迫使两次掉球并拦截了他自己的四次传球。卡罗莱纳的第二防线有职业碗安全卫科特·科尔曼和职业碗角卫约什·诺曼，科尔曼带领球队完成了职业生涯中高达七次拦截并同时贡献了 88 次擒抱，而诺曼在本赛季成长为一名封锁角卫并完成了四次拦截，其中两次被判触地得分。",

2. "华沙自古以来就是一个多元文化的城市。根据 1901 年的人口普查，711,988 名居民中，天主教徒占 56.2%，犹太人占 35.7%，希腊东正教徒占 5%，新教徒占 2.8%。八年后，在 1909 年有 281,754 名犹太人 (36.9%)、18,189 名新教徒 (2.4%) 和 2,818 名玛利亚派 (0.4%)。这导致该镇各地修建了数百个宗教礼拜场所，其中大部分在 1944 年华沙起义后被摧毁。战争结束后，波兰的新共产主义当局不鼓励建造教堂，只有一小部分教堂得以重建。",

3. 在考虑计算问题时，问题实例是字母表上的字符串。通常，字母表被认为是二进制字母表（即，集合{0,1}），因此字符串是位串。就像现实世界中的计算机一样，必须对位串以外的数学对象进行适当的编码。例如，整数可以用二进制记数法表示，图形可以通过其邻接矩阵直接编码，或者通过二进制对其邻接表进行编码。人们很容易认为函数问题的概念要比决策问题的概念丰富得多。然而，实际情况并非如此，因为函数问题可以重新定义为决策问题。例如，两个整数的乘法可以表示为三元组 (a, b, c)，使得关系 a × b = c 成立。确定一个给定的三元组是否属于该集合相当于解决两个数相乘的问题。",

Chinese: AI GENERATED

1. "黑豹队的防守只丢了308分，排名联盟第六，同时以24次拦截领跑NFL，并有四名球员入选职业碗。职业碗防守截锋Kawann Short以11次擒杀领跑全队，同时制造了3次掉球并恢复了2次。同样在防守线上，Mario Addison贡献了6次擒杀。黑豹队的防守线还包括资深的防守端锋Jared Allen，他是五次职业碗球员，也是NFL现役的生涯擒杀领跑者，共有136次擒杀。此外，防守端锋Kony Ealy在仅9次首发中贡献了5次擒杀。在线卫位置上，黑豹队的三名首发线卫中有两人也被选入了职业碗：Thomas Davis和Luke Kuechly。Davis贡献了5次擒杀、制造了4次掉球并有4次拦截，而Kuechly以118次擒抱领跑全队，同时制造了2次掉球并有4次拦截。在后卫线上，职业碗安全卫Kurt Coleman以职业生涯新高的7次拦截领跑全队，并有88次擒抱，职业碗角卫Josh Norman则在赛季中发展为封锁型角卫，贡献了4次拦截，其中2次被返回达阵。"

2. "华沙自古以来就是一个多元文化的城市。根据1901年的人口普查，在711,988名居民中，56.2%是天主教徒，35.7%是犹太人，5%是希腊东正教徒，2.8%是新教徒。八年后的1909年，犹太人有281,754人（占36.9%），新教徒有18,189人（占2.4%），马利亚维特教徒有2,818人（占0.4%）。这促使了在城市各个地区建造了数百个宗教礼拜场所。然而，它们中的大多数在1944年华沙起义后被摧毁。战后，波兰新成立的共产党政府抑制了教堂的建设，只有少数教堂得以重建。"

3. "计算复杂性理论是理论计算机科学中计算理论的一个分支，主要关注根据问题的内在难度对其进行分类，并将这些类别相互关联。所谓的计算问题，指的是原则上可以通过计算机解决的任务，这等同于表示该问题可以通过数学步骤的机械应用，如算法，来解决。在考虑计算问题时，问题实例是一个字母表上的字符串。通常，字母表被认为是二进制字母表（即集合 {0,1}），因此字符串是位串。在现实计算机中，除位串以外的数学对象必须被适当地编码。例如，整数可以用二进制表示，图形可以通过其邻接矩阵直接编码，或者通过将其邻接列表编码为二进制形式。"

Figure 8: Displays the Human (Left) & AI-generated (Right) text in Chinese (Mandarin)

Arabic: Human Written

1.

لم يتخلى فريق بانثرز سوى عن 308 نقطة، ليحتل المركز السادس في الدوري، متصدراً في الوقت عينه دوري كرة القدم الأميركية في الاعتراضات مع 24 اعتراضاً ومتمتعاً بـ أربعة اختيارات في قائمة برو بول. وقد تصدر المدافع المعترض كاوان شورت الفريق في الاستحواذ مع 11 استحواذاً، بينما أجبر الخصم أيضاً على ارتكاب ثلاثة أخطاء واسترجع اثنين. وقد أيضاً المدافع الأخير المخضرم Panthers أضاف زميله لاعب الخط ماريو أديسون 6 استحواذات ونصف. وقد ضم خط جاريد آلان، وهو لاعب كرة قدم اختير 5 مرات في قائمة برو بول وكان متصدر اللاعبين النشطين لدوري كرة القدم الأميركي في الاستحواذ مع 136 استحواذاً، إلى جانب المدافع الأخير كوني إيلي، الذي حصل على 5 استحواذات في 9 الثلاثة الأساسيين للعب في برو بول: توماس ديفيس Panthers بدايات فقط. وخلفهم، تم أيضاً اختيار اثنين من أظهرة فريق و لوك كوتشلي. قام ديفيز بتحصيل 5 استحواذ، وأربعة مرات أجبر الخصم على ارتكاب أخطاء، وأربعة اعتراضات، في حين قاد كوتشلي الفريق في العرقلات (118) وأجبر الخصم على ارتكاب خطأين، واعترض أربع تمريرات بنفسه. وضم ثانوي كارولاينا مؤمن برو بول كورت كولمان، الذي تصدر الفريق بأعلى رقم في مسيرته بسبعة اعتراضات، في حين جمع أيضاً ما يصل إلى 88 عرقلة ومدافع الزاوية في قائمة برو بول جوش نورمان، الذي تحول إلى ركن إقفال خلال الموسم وقام بأربع اعتراضات، تم إرجاع اثنين منها للهدف.

2.

طوال تاريخها، كانت وارسو مدينة متعددة الثقافات. وفقًا لإحصاء 1901، كان من بين 711.988 من السكان 56,2٪ من الكاثوليك و 35,7٪ من اليهود و 5٪ من المسيحيين الأرثوذكس اليونانيين و 2,8٪ من البروتستانت. بعد ثماني سنوات، في عام 1909، كان هناك 281.754 يهودياً (36,9٪) و 18.189 بروتستانت (2,4٪) و 2818 مارياويأ (0,4٪). وقد أدى هذا إلى بناء مئات أماكن العبادة الدينية في جميع أنحاء المدينة. تم تدمير معظمها في أعقاب انتفاضة وارسو في 1944. بعد الحرب، لم تشجع السلطات الشيوعية الجديدة في بولندا بناء الكنائس ولم يتم إعادة بناء سوى عدد قليل منها.",

3.

نظرية التعقيد الحسابي هي فرع من نظرية الحوسبة في علوم الحاسوب النظرية التي تركز على تصنيف المسائل الحاسوبية وفقاً لـ صعوبتها المتأصلة، وربط تلك الأصناف ببعضها البعض. يُفهم أن المسألة الحاسوبية هي مهمة قابلة من حيث المبدأ للحل بواسطة كمبيوتر، وهو ما يعادل القول بأن المشكلة يمكن حلها عن طريق التطبيق الميكانيكي لخطوات رياضية، مثل الخوارزمية. عند التفكير في المسائل الحاسوبية، تكون حالة المسألة عبارة عن سلسلة فوق أبجدية. عادةً ما تؤخذ الأبجدية على أنها الأبجدية الثنائية (أي، المجموعة {0,1})، وبالتالي فإن السلاسل هي سلاسل البت. كما هو الحال في الحاسوب الحقيقي، يجب تشفير العناصر الرياضية بخلاف سلاسل البت بشكل مناسب. فعلى سبيل المثال، يمكن تمثيل الأعداد الصحيحة في الترميز الثنائي، ويمكن تشفير الرسوم البيانية مباشرة عبر المصفوفات المجاورة لها، أو عن طريق ترميز قوائمها المجاورة باللغة الثنائية.",

Arabic: AI GENERATED

1.

"قدمت دفاعات فريق بانثرز أداءً متميزًا حيث سمحت فقط بـ 308 نقاط، مما جعلهم يحتلون المرتبة السادسة في الدوري Pro Bowl. تميز كما قادوا الدوري في الاعتراضات بـ 24 اعتراضًا، وحصل أربعة من لاعبيهم على اختيارات لمباراة الـ ،بكونه قائد الفريق في عدد الإطاحات بـ 11 إطاحة، Pro Bowl لاعب الدفاع كاوان شورت، الذي تم اختياره لمباراة الـ بالإضافة إلى إجباره على ثلاثة أخطاء واسترجاعه لكرتين. وأضاف زميله في الفريق ماريو أديسون 6 إطاحات. كما شملت وكان Pro Bowl تشكيلة الفريق في الخط الدفاعي اللاعب المخضرم جاريد آلين، الذي اختير لخمس مرات في مباراة الـ بـ 136 إطاحة، بالإضافة إلى زميله اللاعب كوني إيلي، الذي حقق 5 إطاحات NFL يحمل لقب قائد الإطاحات النشط في الـ Pro Bowl: في 9 مباريات فقط. خلفهم، تم اختيار اثنين من لاعبي خط الدفاع الثلاثة الأساسيين في الفريق لمباراة الـ توماس ديفيس ولوك كوشلي. حقق ديفيس 5 إطاحات، وأجبر على أربعة أخطاء، واعترض أربعة تمريرات، بينما قاد كوشلي الفريق في عدد التدخلات (118)، وأجبر على خطأين، واعترض أربع تمريرات. كما تميز خط الدفاع الثانوي وقاد الفريق بـ 7 اعتراضات، بالإضافة إلى Pro Bowl للفريق بوجود اللاعب كيرت كولمان، الذي اختير لمباراة الـ تسجيله 88 تدخل، واللاعب جوش نورمان الذي تطور ليصبح مدافعًا لا يمكن تجاوزه خلال الموسم، واعترض أربع تمريرات، أعاد اثنتين منها لتحقيق نقاط."

2.

"كانت وارسو طوال تاريخها مدينة متعددة الثقافات. وفقًا لتعداد عام 1901، من بين 711,988 نسمة كان 56.2% منهم كاثوليك، و35.7% يهود، و5% من المسيحيين الأرثوذكس، و2.8% من البروتستانت. بعد ثماني سنوات، في عام 1909 كان هناك 281,754 يهودي (36.9%)، و18,189 بروتستانتي (2.4%)، و2,818 ماريافيت (0.4%). أدى هذا إلى بناء مئات من أماكن العبادة الدينية في جميع أنحاء المدينة. تم تدمير معظمها بعد انتفاضة وارسو عام 1944. بعد الحرب شجعت السلطات الشيوعية الجديدة في بولندا على عدم بناء الكنائس، وتم إعادة بناء عدد قليل منها."

3.

"نظرية التعقيد الحسابي هي فرع من نظرية الحوسبة في علوم الحاسوب النظرية، التي تركز على تصنيف المشاكل الحسابية وفقًا لصعوبتها الذاتية وربط تلك الفئات ببعضها البعض. تُفهم المشكلة الحسابية على أنها مهمة يمكن من حيث المبدأ حلها بواسطة الكمبيوتر، وهو ما يعادل القول بأن المشكلة يمكن حلها بالتطبيق الميكانيكي للخطوات الرياضية، مثل الخوارزمية. عند النظر في المشاكل الحسابية، فإن مثيل المشكلة هو سلسلة من الرموز على أبجدية. عادةً، تُعتبر الأبجدية هي الأبجدية الثنائية (أي المجموعة {0,1})، وبالتالي تكون السلاسل سلاسل بت. كما هو الحال في الكمبيوتر الواقعي، يجب ترميز الأشياء الرياضية غير سلاسل البت بشكل مناسب. على سبيل المثال، يمكن تمثيل الأعداد الصحيحة في الترميز الثنائي، ويمكن ترميز الرسوم البيانية مباشرة عبر مصفوفاتها المجاورة، أو بترميز قوائم الجوار في صورة ثنائية."

Figure 9: Displays the Human (Left) & AI-generated (Right) text in Arabic

1. **NFL Team – Carolina Panthers' Description**
   - Can you provide a detailed breakdown of the Carolina Panthers' defensive stats for their standout season 2015?
     - Specifically mention how many points they allowed and where they ranked in the NFL.
     - Include how many interceptions the team had and who led the team in sacks.
     - Mention Pro Bowl selections, especially defensive tackle Kawann Short's contributions and Mario Addison's sack total.
     - Also mention Jared Allen's career sack total and Kony Ealy's stats.
     - Highlight the linebackers' performance, especially Thomas Davis and Luke Kuechly.
     - Include details on Pro Bowl selections in the secondary, such as Kurt Coleman's interceptions and Josh Norman's defensive stats.

2. **Warsaw's Multicultural History**
   - Can you summarise Warsaw's population demographics from the 1901 census?
     - Include a breakdown of the different religious groups by percentage.
     - Provide an update on how the population changed by 1909, including specific numbers of Jews, Protestants, and Mariavites.
     - Mention how this multicultural population influenced the construction of places of worship.
     - Provide details on what happened to these places of worship during the Warsaw Uprising and how post-war communist authorities handled rebuilding efforts.

3. **Computational Complexity Theory**
   - Can you explain the concept of computational complexity theory in theoretical computer science?
     - Define what a computational problem is and how it relates to algorithms.
     - Describe how instances of computational problems are represented as strings made up of symbols from an alphabet.
     - Specifically mention the use of binary digits in these representations.
     - Explain how other mathematical entities like integers and graphs are encoded for computational purposes.
     - Provide examples of how integers are represented in binary form and how graphs are encoded using adjacency matrices or lists.

Figure 10: Questions used to create Information Paragraphs from GPT-4 for Testing

	TP	TN	FN	FP	UFP	UTP	UTN	UFN	NUC	Accuracy	Accuracy (%)
DetectGPT	5	9	5	5	1	3	0	2	0	0.583	58.333
Go Winston	8	3	2	9	1	3	0	0	4	0.500	50.000
GPT Zero	4	8	6	1	0	2	3	2	4	0.632	63.158
GPT-2 Output Detector Demo	2	13	11	1	0	0	1	1	1	0.556	55.556
Writer	0	13	14	1	0	0	1	1	0	0.464	46.429
ZeroGPT	12	1	1	8	2	2	1	0	3	0.591	59.091

Table 2: Classification of results and accuracy for each AI detector

	TP	TN	FN	FP	UFP	UTP	UTN	UFN	NUC	Accuracy	Accuracy (%)
English	9	10	6	6	0	3	0	0	2	0.613	61.290
Spanish	6	12	9	4	0	0	2	1	2	0.581	58.065
German	5	13	11	3	1	1	0	1	1	0.563	56.250
Chinese	7	7	7	7	1	3	1	1	2	0.500	50.000
Arabic	4	5	6	5	2	3	3	3	5	0.450	45.000

Table 3: Classification of results and accuracy for each language according to detector results

Language (Writer)	DetectGPT	Go Winston	GPT Zero	GPT-2 Output Detector Demo	Writer	ZeroGPT
English Hum-1 (Text 1)	100% (FP)	48% AI (NUC)	4% (TN)	0.04% (TN)	5% (TN)	96.41% (FP)
English Hum-1 (Text 2)	82% (FP)	99% (FP)	5% (TN)	0.03 (TN)	0% (TN)	100% (FP)
English Hum-1 (Text 3)	0% (TN)	99% (FP)	3% (TN)	0.02 (TN)	8% (TN)	50% (NUC)
English AI-2 (Text 1)	76% (UTP)	61% (UTP)	95% (TP)	1.4% (FN)	8% (FN)	100% (TP)
English AI-2 (Text 2)	84% (TP)	100% (TP)	76% (UTP)	80.43 (TP)	16% (FN)	100% (TP)
English AI-2 (Text 3)	100% (TP)	100% (TP)	95% (TP)	0.05% (FN)	4% (FN)	0% (FN)
	DetectGPT	Go Winston	GPT Zero	GPT-2 Output Detector Demo	Writer	ZeroGPT
Spanish Hum-1 (Text 1)	0% (TN)	0% (TN)	0% (TN)	0.03 (TN)	3% (TN)	95.4% (FP)
Spanish Hum-1 (Text 2)	0% (TN)	0% (TN)	31% (UTN)	0.02 (TN)	2% (TN)	100% (FP)
Spanish Hum-1 (Text 3)	0% (TN)	99% (FP)	31% (UTN)	0.02 (TN)	1% (TN)	100% (FP)
Spanish AI-2 (Text 1)	100% (TP)	59% (NUC)	54% (NUC)	0.03 (FN)	1% (FN)	94.69 (TP)
Spanish AI-2 (Text 2)	0% (FN)	0% (FN)	100% (TP)	0.02 (FN)	6% (FN)	100% (TP)
Spanish AI-2 (Text 3)	20% (UFN)	0% (FN)	99% (TP)	0.02(FN)	1% (FN)	80.75% (TP)
	DetectGPT	Go Winston	GPT Zero	GPT-2 Output Detector Demo	Writer	ZeroGPT
German Hum-1 (Text 1)	0% (TN)	90% (FP)	1% (TN)	0.03 (TN)	0% (TN)	0% (TN)
German Hum-1 (Text 2)	0% (TN)	50% (NUC)	0% (TN)	0.02 (TN)	3% (TN)	78.5% (UFP)
German Hum-1 (Text 3)	0% (TN)	0% (TN)	90% (FP)	0.01 (TN)	0% (TN)	100% (FP)
German AI-2 (Text 1)	0% (FN)	60% (UTP)	4% (FN)	0.05% (FN)	10% (FN)	95.8% (TP)
German AI-2 (Text 2)	0% (FN)	81% (TP)	3% (FN)	0.03 (FN)	16% (FN)	100% (TP)
German AI-2 (Text 3)	20% (FN)	100% (TP)	0% (FN)	0.9% (FN)	20% (UFN)	99% (TP)

Table 10: Results of AI Detection Tools for Latin-Based Languages

	DetectGPT	Go Winston	GPT Zero	GPT-2 Output Detector Demo	Writer	ZeroGPT
Chinese Hum-1 (Text 1)	99% (FP)	95% (FP)	4% (TN)	4% (TN)	7% (TN)	56.6% (NUC)
Chinese Hum-1 (Text 2)	95% (FP)	100% (FP)	15% (TN)	0.95% (TN)	28% (UTN)	79% (UFP)
Chinese Hum-1 (Text 3)	0% (TN)	99% (FP)	41% (NUC)	1% (TN)	100% (FP)	100% (FP)
Chinese AI-2 (Text 1)	0% (FN)	79% (UTP)	10% (FN)	0.2% (FN)	8% (FN)	98% (TP)
Chinese AI-2 (Text 2)	85% (TP)	90% (TP)	26% (UFN)	80.9% (TP)	7% (FN)	100% (TP)
Chinese AI-2 (Text 3)	78% (UTP)	100% (TP)	68% (UTP)	0.59 (FN)	16% (FN)	85.69 (TP)
	DetectGPT	Go Winston	GPT Zero	GPT-2 Output Detector Demo	Writer	ZeroGPT
Arabic Hum-1 (Text 1)	0% (TN)	92% (FP)	34% (UTN)	94.43 (FP)	1% (TN)	36.75% (UTN)
Arabic Hum-1 (Text 2)	87% (FP)	99% (FP)	58% (NUC)	39.5 (UTN)	3% (TN)	45.9% (NUC)
Arabic Hum-1 (Text 3)	69% (UFP)	78% (UFP)	55% (NUC)	13.65 (TN)	4% (TN)	90% (FP)
Arabic AI-2 (Text 1)	79% (UTP)	100% (TP)	4% (FN)	47.17% (NUC)	0% (FN)	79.9% (UTP)
Arabic AI-2 (Text 2)	84% (TP)	59% (NUC)	14% (FN)	3.18 (FN)	3% (FN)	100%% (TP)
Arabic AI-2 (Text 3)	25% (UFN)	98% (TP)	33% (UFN)	23.16% (UFN)	4% (FN)	66.9% (UTP)

Table 11: Results of AI Detection Tools for non-Latin-Based Languages